

Mahbub H. Raton

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EDUCATION

University of California, Irvine

PhD in Computer Science

Irvine, CA

Expected graduation: 04/28

- **Advisor:** Prof. Michael Franz, Secure Systems and Software Lab.
- **Focus:** Compilers and computer architecture: compiler-assisted detection of transient hardware errors via LLVM transformation passes, and cross-architecture (x86-to-AArch64) binary translation.

University of California, Irvine

Master of Science in Computer Science (en route to PhD)

Irvine, CA

Conferred: 06/26

Bangladesh University of Engineering & Technology

Bachelor of Science in Computer Science & Engineering (GPA: 3.71)

Dhaka, Bangladesh

Conferred: 02/21

RESEARCH EXPERIENCE

University of California, Irvine

Graduate Research Assistant — Secure Systems and Software Lab (Advisor: Prof. Michael Franz) Sep 2023 – Present

Irvine, CA

- Doctoral research in compilers, computer architecture, and systems reliability, developing solutions for real-world, large-scale systems-reliability problems (LLVM, gem5).

AEGIS: LLVM Compiler Pass for Hardware Reliability | C++, LLVM, Clang, CMake Aug 2025 – Present

- Implemented Error Detection by Duplicated Instructions (EDDI) as an LLVM pass that duplicates global, stack, and heap values and compares them before stores, branches, and function calls to detect corruption.
- Modified function call sites to pass shadow arguments and retrieve shadow return values; resolved pointer sizes via DWARF debug metadata and a library signature table.
- Designed an extensible strategy registry (software design for maintainability); adding a new hardening scheme requires a single self-registering file.
- Wrote regression tests validating pass correctness and fault-detection coverage across the benchmark suite.
- Reduced silent data corruption by $1.3\times$ – $29\times$ across `qsort`, `matmul`, and `CRC32` benchmarks under L1 cache fault injection. Runtime performance overhead stayed at $1.6\times$ – $3.0\times$.

Architectural Fault Injection Framework | C++, Python, gem5, Docker, Linux

Aug 2025 – Present

- Developed Python scripts to automate a large-scale fault-injection campaign, using gem5 checkpointing and distributed multi-machine parallelization; accelerated a 600K-injection campaign across three benchmarks from 7 days to under 24 hours.
- Added register-file and L1 cache fault injection to gem5 in C++, pinpointing the first instruction that reads corrupted data at under 0.1% simulation overhead.
- Automation harness runs injections in parallel and classifies outcomes (SDC, masked, detected, crash, timeout) with 99% confidence margins; finds root cause by diffing execution traces.
- **Source:** github.com/mahbub-hr/gem5

Cross-Compiling x86 Binaries to ARM | LLVM, C++, S2E, x86/ARM Assembly

Jun 2024 – Jun 2025

- Extended the `Binrec` framework to enable x86-to-ARM binary translation by adapting its dynamic execution and LLVM IR lifting pipeline.
- Implemented an x64-to-AArch64 translation layer for library calls; validated correct execution on two real-world SPEC CPU 2017 programs, `mcf` and `xalancbmk` ($6\times$ slowdown relative to native).

PUBLICATIONS & RESEARCH COMMUNITY SERVICE

Research Manuscripts: M. H. Raton, M. Franz. “*Architectural Hurdles: Exploring the Difficulties of x86-to-ARM Binary Cross-Compilation*” (Manuscript in Preparation).

M. H. Raton, A. Abdullah. “*SliveredChain: Reducing Storage in Private Blockchain Systems*” (Undergraduate Thesis; BFT consensus over a sharded design, reducing storage by 60%).

Research Community Service: Artifact Evaluation Committee: EuroSys 2026, PACT 2026. Reviewer: IEEE ICMLA 2024, ACM SAC 2026.

SELECTED PROJECTS

- IR-Sentinel: LLVM-IR Security Analysis via MCP | *Python, LLVM, Gemini, FastMCP*

Feb 2026

 - Developed a proof-of-concept **Model Context Protocol (MCP)** server (**FastMCP**) exposing local LLVM toolchains (**clang**, **opt**) to AI agents.
 - The agent pipeline compiles C++ to **LLVM IR**, extracts **CFG** metrics, and iteratively queries **Gemini** to identify Use-After-Free vulnerability patterns.
 - Source:** github.com/mahbub-hr/ir-sentinel
- WARIR: Wasm Optimizer Compiler (graduate compiler course project) | *Python, Node.js, WebAssembly*

Jan 2024 – Jun 2024

 - Implemented **graph-coloring register allocation** and **SSA-based dataflow analysis** (Common Subexpression Elimination, Copy Propagation) as part of a full compiler pipeline (SMPL-to-WebAssembly).
 - Source:** github.com/mahbub-hr/compiler_cs241

TEACHING & INDUSTRY EXPERIENCE

- University of California, Irvine

Irvine, CA

Graduate Teaching Assistant

Sep 2023 – Present

 - Mentored 300+ students across 6 CS courses. Assisted instructors with grading and question design.
- United International University

Dhaka, Bangladesh

Lecturer in Computer Science

Dec 2021 – Jul 2023

 - Delivered lectures for “Object Oriented Programming” and “Pattern Recognition” (Avg class size: 45).
 - Co-advised senior thesis projects on Data Mining and collaborated on syllabus modernization.
- Banbeis (Ministry of Education)

Dhaka, Bangladesh

Software Developer

Mar 2021 – Dec 2021

 - Engineered the backend (**Spring Boot**, **Laravel**) for a nationwide production high school admission system (CAD) serving **1.3 million** student applicants across 4 education boards.
 - Designed the schema and wrote the SQL Server stored procedures for post-deadline merit ranking and school migrations across 4 education boards.
 - Diagnosed a **GC overhead limit exceeded** OutOfMemoryError in the 1.3M-applicant ranking query executed via Spring Data; resolved it by replacing entity-based retrieval with the JDBC API.
 - Collaborated within a 5-person engineering team and with cross-functional module teams to expose data via REST APIs secured with Keycloak (OAuth2); deployed through Jenkins CI/CD pipelines and participated in deployment planning meetings.

TECHNICAL SKILLS

- Programming:** C, C++ (LLVM Pass Dev), Python (Advanced), Java, JavaScript, x86 & ARM Assembly, CUDA, Shell Scripting.
- Compilers & Architecture:** LLVM Framework, Gem5, GDB, SVF (Static Value Flow), Control Flow Graphs (CFG).
- Data Structures, Algorithms & Software Design:** Graph Algorithms, Register Allocation, SSA-Based Dataflow Analysis, Merkle Trees, Distributed Systems (BFT, Paxos Consensus, Sharding).
- Backend & DB:** Flask, Node.js, SQL (MySQL/SQLite/SQL Server), REST APIs, Keycloak (OAuth2/OIDC).
- AI Systems & Tooling:** Model Context Protocol (MCP), Google GenAI (Gemini), OpenAI API.
- DevOps & Tools:** Git, GitHub, Docker, Jenkins (CI/CD), JIRA.

AWARDS

- Dean’s Fellowship:** UC Irvine (2024-2025).
- Dean’s List & Merit Scholarship:** BUET (2016-2020).